# ECSE 4470, MANE 4550 - Intro to Robot Programming

**Basic Overview** 1
- Course Instructor . . . 1
- Office Hours . . . 1
- Teaching Assistant . . . 1
- Class Schedule and Location . . . 1
- Catalog Description . . . 1
- Student Learning Outcomes . . . 1
- Major Topics Covered . . . 1
- Pre-Requisite Courses . . . 2
- Texts . . . 2

**Performance Assessment** 3
- Grade Composition . . . 3
- Numerical to Letter Grade Conversion . . . 3
- Labs . . . 3
- Final Project . . . 4
- Quizzes . . . 4
- ICF . . . 4
- Late Submissions . . . 5

**Course Policies** 6
- Syllabus or Course Changes . . . 6
- Attendance . . . 6
- Lab Etiquette . . . 6
- Grade Appeals . . . 6
- Late Submissions . . . 6
- Academic Accommodations . . . 6
- Diversity, Equity, and Inclusivity . . . 6
- AI Tool Usage . . . 7
- Digital Tool Acknowledgment . . . 7
- Statement of Academic Integrity . . . 7

**Course Outline** 9

# Basic Overview

**Course Instructor:** Kimberly Oakes, JEC 7006, oakesk2@rpi.edu

**Office Hours:** TBD

**Teaching Assistant:** TBD

**Class Schedule and Location:**

- Tues/Fri 12:30-1:50pm Eastern Time
- TBD
  - Some lectures will will be recorded and available on the course Piazza site
- Online Course Spaces
  - LMS (Grades only): https://lms.rpi.edu
  - Piazza (Materials): https://piazza.com/rpi/spring2025/ecse4965mane4961/resources
  - WebEx Team (Communication): Instructor will add students
  - Gradescope (Assignment submission): https://www.gradescope.com
  - Notion (Materials being migrated to here): https://robot-programming-course.notion.site/
- Recorded Sessions: All lectures will be delivered during the class hours and some will be recorded using WebEx and made available through Piazza.

**Catalog Description:** This course provides an introduction to Robot Operating System (ROS), a widely used middleware for robot programming. Students will learn the basic components of ROS, how to utilize and write ROS packages, gain experience utilizing simulation software, and apply the concepts to physical platforms. *3 credit hours*

**Student Learning Outcomes:** Students who complete this course satisfactorily will be able to:

(1) Apply the basics of Python programming to create ROS communication capable code

(2) Identify ROS components such as nodes, topics, services, actions, launch files, etc. and explain how they differ between ROS1 and ROS2

(3) Apply ROS components knowledge to write and utilize ROS packages to achieve a specific task

(4) Successfully test developed code using simulation tools (Rviz, Gazebo)

**Student Learning Outcomes Assessment Measures:**

**Major Topics Covered:** Students who complete this course satisfactorily will be able to understand and apply the following: (1) Linux system command line usage (2) ROS components (3) ROS packages (4) URDF files (5) Rviz (6) MoveIt! (7) Gazebo (8) ROS1 vs ROS2 (9) Application to real systems

| Assessment | Learning Outcome #s |
|---|---|
| Quizzes | 3 |
| Lab Reports | 1,2,4,5 |
| Final Project | 1,2,3,4,5 |

**Pre-Requisite Courses:**

- CSCI-1100 (Computer Science I)

**Texts:** None. The course will use instructor notes and online materials.

**Supplements:**

- J. M. O'Kane, *A Gentle Introduction to ROS*. Columbia, SC: Jason M. O'Kane, 2014.
- https://pythonbooks.org/free-books/

# Performance Assessment

**Grade Composition:**

| Component | Weight |
|---|---|
| Quizzes | 10% |
| Lab Reports | 55% |
| Final Project | 35% |
| Total | 100 % |

| Final Project Component | Weight |
|---|---|
| Presentation | 40% |
| Report | 60% |
| Total | 100 % |

**Numerical to Letter Grade Conversion:** Numerical to letter grade mapping is as follows:

| Numeric Grade | Letter Grade |
|---|---|
| $\geq 93$ | A |
| $\geq 89.67$ | A- |
| $\geq 86.33$ | B+ |
| $\geq 83$ | B |
| $\geq 79.67$ | B- |
| $\geq 76.33$ | C+ |
| $\geq 73$ | C |
| $\geq 69.67$ | C- |
| $\geq 66.34$ | D+ |
| $\geq 63$ | D |
| $< 63$ | F |

**Labs:** The labs will emphasize application of the various topics explored in the lectures and quizzes. Each lab submission will consist of a technical report. This report should be written in a professional manner (neatness and thoroughness count!). They should be submitted as a single pdf file on Gradescope. Program code and Read Me file should be put on GitHub or Google Drive with link included in your report. The code should be self-contained so that the instructor can run the code for evaluation and well commented.

The grade of the report is roughly distributed as follows: Results 50%, Approach 30%, Presentation/organization/writing 20%. Please see the full report grading rubric for more information.

Your report should use the following outline:

Report Outline

- Hightlights
  - Include bulleted list
- Problem(s) Encountered / Solution(s) Found
  - Problem #
    - Solution
      - Debug Details
- Details of Execution
  - What you did, how, and **why**
- Future Directions. Possibly including:
  - What to explore in more depth?
  - What other skills/knowledge would be helpful?
- External References Utilized

| Lab Number | Topics | Start Date | Due Date |
|---|---|---|---|
| 1 | VM installation; Linux command line | 1/13 | 1/30 |
| 2 | Turtlesim – ROS basics | 1/23 | 2/6 |
| 3 | Write ROS node/package (camera) | 2/3 | 2/24 |
| 4 | Turtlesim – Advanced topics | 2/24 | 3/13 |
| 5 | Application of ROS2 to RaspbotV2 | 3/10 | 3/27 |
| 6 | Writing URDF, Using Rviz | 3/27 | 4/7 |
| 7 | Using Gazebo | 4/7 | 4/17 |

**Final Project:** The final project will be a demonstration of the application of multiple course topics to achieve some uniquely defined task. This is an open-ended project to be defined by the students. Projects will be completed in teams. Grading will address general communication skills, thoroughness and accuracy of presented material, and quality and innovation of the final demonstration. Deliverables consist of an oral presentation and written report. The report should use the same outline as the labs. It is expected that the final report will include more details and references than the lab reports.

**Quizzes:** Quizzes will be posted in Gradescope every 1-2 weeks. Material will emphasize topics covered within the last 1-2 weeks but may address previous topics as well. Quizzes must be done individually without collaboration or discussion until after submission and grading is complete.

**ICF:** For each lab and the final project, feedback forms will be requested from all team members. If a team member is not sufficiently contributing to the teamwork, they will receive a scaled version of the team's grade. Repeated low ICFs will result in removal from the team, requiring the individual to work independently. Any concerns or factors that will impact your ability to contribute should be communicated to your team and/or instructor as appropriate.

| Quiz Number | Topics | Start Date | Due Date |
|---|---|---|---|
| 1 | Linux terminal | 1/13 | 1/20 |
| 2 | ROS components | 1/20 | 1/27 |
| 3 | Python | 2/3 | 2/10 |
| 4 | ROS nodes and packages | 2/10 | 2/20 |
| 5 | URDF files, Rviz, MoveIt! | 3/20 | 3/27 |
| 6 | Cummulative Review | 3/27 | 3/31 |
| 7 | Gazebo | 4/7 | 4/17 |
| 8 | ROS1 vs ROS2 | 4/17 | 4/28 |

**Late Submissions:** Assignments submitted on Gradescope have a submission due date and time. It may also have a late deadline. This late deadline can be used without penalty. Submissions will not be accepted after the late deadline unless there has been prior communication (at least 24hour notice of the original deadline) with the instructor.

# Course Policies

**Syllabus or Course Changes:** The outline of topics and assignments may be subject to change. All changes will be communicated as soon as possible. Any concerns should be addressed to the instructor as soon as possible.

**Attendance:** Attendance is expected in every class period, unless previously discussed with the instructor, and if necessary, officially documented by the Student Experience office (4th floor, Academy Hall).

**Lab Etiquette:** Card access will be given to JEC 5312 to work on your projects. Students will be held to high standards with regards to cleanliness, professionalism, and consideration. Food and drink are not allowed in the lab space. The equipment is new and sensitive. Damage resulting from spilled food or drink will result in revocation of lab privileges. Similarly, if lab cleanliness overall becomes a problem, lab hours may be restricted or revoked for those deemed responsible. Managing shared work spaces and resources is an important skill for all professionals. Any concerns regarding lab etiquette should be brought to the attention of the instructor as soon as possible.

**Grade Appeals:** Requests for regrade must be made within 24hours of grade being communicated to the student. Any concerns regarding overall course performance must be communicated to the instructor as soon as possible. Final grades will not be modified once published in SIS unless there is a demonstrable error made by the instructor (i.e. a calculation error).

**Late Submissions:** Assignments submitted on Gradescope have a submission due date and time. It may also have a late deadline. This late deadline can be used without penalty. Submissions will not be accepted after the late deadline unless there has been prior communication (at least 24hour notice of the original deadline) with the instructor.

**Academic Accommodations:** Rensselaer Polytechnic Institute is committed to providing equal access to our educational programs and services for students with disabilities. If you anticipate or experience academic barriers due to a disability, please contact the Office of Disability Services for Students (DSS) (dss@rpi.edu; 518-276-8197) to establish reasonable accommodations. Once you have been approved for accommodations, please provide your Faculty Memorandum (a letter provided to students by DSS) to all faculty members for this course. Please provide this at the beginning of the semester.

**Diversity, Equity, and Inclusivity:** The goal of the instructor is to provide an environment that promotes learning and success for students of all backgrounds and learning styles. The ability to collaborate and support each other when working in teams is a very important life skill. All students are expected to demonstrate respect for their peers in and out of the classroom. Discrimination based on gender identity, race, ethnicity, class, religion, sexuality, disability, etc. is not acceptable.

Any concerns regarding the pacing or structure of the course should be communicated to the instructor. To accommodate different learning styles, I have taken some steps such as the provision of lecture notes and recordings as well as the use of dyslexic friendly fonts. Other suggestions are welcome.

School of Engineering policy is that no students, faculty and staff shall be excluded on the grounds of gender, race, ethnicity, class, religion, sexuality, disability, etc. For more information on RPI's approach to promoting diversity, equity, and inclusion, please see https://info.rpi.edu/diversity

**AI Tool Usage:** AI tools can be useful for helping to understand concepts or rewording questions for possible clarification. Usage of AI tools to complete homeworks must be acknowledged. Direct copying of the outputs is not allowed. Submissions must be rewritten in your own words.
**!!!AI TOOLS MAY NOT BE USED TO COMPLETE EXAMS!!!** The use of AI tools to complete the exams will result in a zero for that exam. Repeated violation will result in a report of academic dishonesty to the Dean of Students for possible additional actions.

**Digital Tool Acknowledgment:** As described in the syllabus, Robotics I uses several external services to manage the content and grading of the course, including Piazza, Gradescope and Webex Teams. These online tools provide services designed to assist schools, teachers, and other educational partners in improving student learning outcomes. In some circumstances, these online tools may receive personally identifiable information about students ("Student Data") from the instructor in the course providing this service. For example, an instructor will provide a class roster, email addresses of all students in the class, as well as coursework data that may be linked to a particular student. All listed online resource companies used by the instructor consider Student Data to be strictly confidential and have physical, administrative, and technical security protections in place to protect such data. They do not use personally identifiable Student Data for any purpose other than to provide the services to the instructor, and they do not share personally identifiable Student Data with any third party except as authorized or required by the instructor. The online tools above may collect, analyze, and share anonymized or aggregated data or data derived from Student Data for certain purposes, but only if the disclosure of such data could not reasonably identify a specific individual or specific School. The collection and use of Student Data provided by the instructor is governed by the Terms of Service for each platform and by the provisions of the Family Educational Rights and Privacy Act (FERPA). Student data is provided and controlled by the instructor. If you have questions about reviewing, modifying, or deleting your personal information, please contact the instructor. All live sessions will be recorded, including your questions and comments and possibly your video if your camera is on. If you do not want to be part of the recordings, please be sure to turn off your camera and microphone. By agreeing to this policy (via Gradescope), you agree to the usage of the digital tools mentioned above for distributing, collecting, and grading your homework, exams, and chats in this course this semester. Your assignments will not be graded without you agreeing to this policy. Please see the instructor if you have any concerns about how a specific tool will be used.

**Statement of Academic Integrity:** Student-teacher relationships are built on trust. For example, students must trust that teachers have made appropriate decisions about the structure and content of the courses they teach, and teachers must trust that the assignments that students turn in are their own. Acts that violate this trust undermine the educational process. The Rensselaer Handbook of Student Rights and Responsibilities and Graduate Student Supplement define various forms of Academic Dishonesty. Familiarize yourself with these definitions. In this class, all assignments that are turned in for a grade must represent the student's own work.

Three relevant academic integrity violations to emphasize include:

• Unallowed Collaboration: Collaboration is defined as deliberately facilitating an act of academic dishonesty in any way or form; for example, allowing another student to observe an exam paper or allowing another student to "recycle" one's old term paper or using one another's work in a paper or lab report without citing it as another's work.

• Copying: Copying is defined as obtaining information pertaining to a graded exercise by deliberately observing the paper of another student; for example, noting which alternative a neighboring student has circled on a multiple-choice exam.

• Plagiarism: Plagiarism is defined as representing the work or words of another as one's own through the omission of acknowledgment or reference. Examples include using sentences verbatim from a published source in a term paper without appropriate referencing, or presenting as one's own the detailed argument of a published source, or presenting as one's own electronically or digitally enhanced graphic representations from any form of media.

Homework, quizzes, and exam submissions must represent the student's own work. **In cases where help was received or given, a notation should be included to indicate the collaboration.** Usage of materials (books, internet sites, etc.) outside the listed resources should be cited. Evidence of academic dishonesty will result in a zero for the associate assignment and a report to the Dean of Students for possible additional actions.

# Course Outline

Notation: C=Chapter, A=Appendix

| Lec | Date | Topic | References | Assignment |
|---|---|---|---|---|
| 1 | 1/13(Tue) | Introduction<br>What is ROS? (main concept) | Overview | - Lab 1 Start<br>∘ Quiz 1 Start |
| 2 | 1/16(Fri) | Linux overview<br>ROS structure overview | Linux Commands | |
| 3 | 1/20(Tue) | Github overview<br>ROS components intro | Github<br>Git Commands | ∗ Quiz 1 Due<br>∘ Quiz 2 Start |
| 4 | 1/23(Fri) | ROS components:<br>nodes, messages, topics, services, actions<br>Utilizing ROS:<br>rqtgraph, rosbag, connectivity, parameters | Humble Docs | - Lab 2 Start |
| 5 | 1/27(Tue) | Python overview<br>Object-Oriented Programming (OOP) in Python | Python tutorials<br>OOP in Python<br>Classes<br>Method Types<br>Class Connections<br>Properties<br>Pythonic Style<br>F-Strings | ∗ Quiz 2 Due |
| 6 | 1/30(Fri) | In-class lab work (turtlesim) | | ! Lab 1 Due |
| 7 | 2/3(Tue) | Introduction to Packages<br>Publishers and Subscribers | | - Lab 3 Start<br>∘ Quiz 3 Start |
| 8 | 2/6(Fri) | Creating Services<br>Custom Interfaces | Interfaces | ! Lab 2 Due |
| 9 | 2/10(Tue) | Creating Actions | async in Python<br>enum | ∗ Quiz 3 Due<br>∘ Quiz 4 Start |
| 10 | 2/13(Fri) | In-class lab work | | |
| | 2/17(Tue) | President's Day (No class) | | |
| 11 | 2/20(Fri) | Some useful tools<br>Parameters, Namespaces, Remapping<br>Writing and using launch files | ros-args<br>Launch files - Python<br>Launch files - all | ∗ Quiz 4 Due |
| 12 | 2/24(Tue) | tf2 | | ! Lab 3 Due<br>- Lab 4 Start |
| 13 | 2/27(Fri) | In-class lab work | | |
| | | **Spring Break** | | |
| 14 | 3/10(Tue) | Utilizing ROS2 on RaspbotV2 | Yahboom | - Lab 5 Start |
| 15 | 3/13(Fri) | In-class lab work | | ! Lab 4 Due |
| 16 | 3/17(Tue) | In-class lab work | | |
| 17 | 3/20(Fri) | In-class lab work | | ∘ Quiz 5 Start |
| 18 | 3/24(Tue) | URDF files | XML Tutorial<br>URDF - ROS Wiki<br>URDF - Example | |

| Lec | Date | Topic | References | Assignment |
|---|---|---|---|---|
| 19 | 3/27(Fri) | In-class lab work | | ! Lab 5 Due<br>- Lab 6 Start<br>∗ Quiz 5 Due<br>∘ Quiz 6 Start |
| 20 | 3/31(Tue) | Rviz<br>MoveIt | Rviz (*Humble distro)<br>MoveIt Tutorials<br>MoveIt Methods | |
| 21 | 4/3(Fri) | In-class lab work | | ∗ Quiz 6 Due |
| 22 | 4/7(Tue) | Gazebo | Gazebo Tips<br>Gazebo Tag | ! Lab 6 Due<br>- Lab 7 Start<br>∘ Quiz 7 Start |
| 23 | 4/10(Fri) | Project work | | |
| 24 | 4/14(Tue) | Project work | | |
| 25 | 4/17(Fri) | Project work | | ! Lab 7 Due<br>∗ Quiz 7 Due<br>∘ Quiz 8 Start |
| 26 | 4/21(Tue) | Project work | | |
| 27 | 4/24(Fri) | Project work | | |
| 28 | 4/28(Tue) | Final presentations | | ∗ Quiz 8 Due |